

1 sel_block.vhd

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-- Title       : sel_block
-- Design      : NIMPlus
-- Author      : Unknown
-- Company     : Unknown
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-- File        : C:\AGP_2018_05_02_NIMPlus_T_C_RJ45\NIMPlus\NIMPlus\compile\se
l_block.vhd
-- Generated   : Fri May 11 15:26:52 2018
-- From        : C:\AGP_2018_05_02_NIMPlus_T_C_RJ45\NIMPlus\NIMPlus\src\sel_bl
ock.bde
-- By          : Bde2Vhdl ver. 2.6
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-- Description :
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-- Design unit header --
library IEEE;
use IEEE.std_logic_1164.all;
use IEEE.std_logic_arith.all;
use IEEE.std_logic_signed.all;
use IEEE.std_logic_unsigned.all;
--use nim_plus_package_1.all;

entity sel_block is
    port(
        blk_en : in STD_LOGIC;
        clk0   : in STD_LOGIC;
        rst_p  : in STD_LOGIC;
        en_term : in STD_LOGIC_VECTOR(15 downto 0);
        x      : in STD_LOGIC_VECTOR(3 downto 0);
        sig_out : out STD_LOGIC;
        logterm : out STD_LOGIC_VECTOR(15 downto 0)
    );
end sel_block;

architecture sel_block of sel_block is

    ---- Component declarations ----

    component d_ff
        port (
            clk : in STD_LOGIC;
            dl  : in STD_LOGIC;
            rst_p : in STD_LOGIC;
            q : out STD_LOGIC
        );
    end component;

    ---- Constants ----

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constant GND_CONSTANT    : STD_LOGIC := '0';

---- Signal declarations used on the diagram ----

signal GND : STD_LOGIC;
signal rst_n : STD_LOGIC;
signal sig_term : STD_LOGIC;
signal sig_un : STD_LOGIC;
signal mterm : STD_LOGIC_VECTOR(3 downto 0);
signal raw_term : STD_LOGIC_VECTOR(15 downto 0);
signal term : STD_LOGIC_VECTOR(15 downto 0);
signal xn : STD_LOGIC_VECTOR(3 downto 0);

begin

---- Component instantiations ----

rst_n <= not(rst_p);

term(2) <= en_term(2) and raw_term(2);

raw_term(3) <= xn(3) and xn(2) and x(1) and x(0);

term(3) <= en_term(3) and raw_term(3);

raw_term(4) <= xn(3) and x(2) and xn(1) and xn(0);

term(4) <= en_term(4) and raw_term(4);

raw_term(5) <= xn(3) and x(2) and xn(1) and x(0);

term(5) <= en_term(5) and raw_term(5);

raw_term(6) <= xn(3) and x(2) and x(1) and xn(0);

term(6) <= en_term(6) and raw_term(6);

raw_term(7) <= xn(3) and x(2) and x(1) and x(0);

xn(0) <= not(rst_n and x(0));

term(7) <= en_term(7) and raw_term(7);

raw_term(8) <= x(3) and xn(2) and xn(1) and xn(0);

term(8) <= en_term(8) and raw_term(8);

raw_term(9) <= x(3) and xn(2) and xn(1) and x(0);

term(9) <= en_term(9) and raw_term(9);

raw_term(10) <= x(3) and xn(2) and x(1) and xn(0);

term(10) <= en_term(10) and raw_term(10);

raw_term(11) <= x(3) and xn(2) and x(1) and x(0);

term(11) <= en_term(11) and raw_term(11);

raw_term(12) <= x(3) and x(2) and xn(1) and xn(0);

xn(1) <= not(rst_n and x(1));

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term(12) <= en_term(12) and raw_term(12);
raw_term(13) <= x(3) and x(2) and xn(1) and x(0);
term(13) <= en_term(13) and raw_term(13);
raw_term(14) <= x(3) and x(2) and x(1) and xn(0);
term(14) <= en_term(14) and raw_term(14);
raw_term(15) <= x(3) and x(2) and x(1) and x(0);
term(15) <= en_term(15) and raw_term(15);
mterm(0) <= term(3) or term(2) or term(1) or term(0);
mterm(1) <= term(7) or term(6) or term(5) or term(4);
mterm(2) <= term(11) or term(10) or term(9) or term(8);
xn(2) <= not(rst_n and x(2));
mterm(3) <= term(15) or term(14) or term(13) or term(12);
sig_term <= mterm(3) or mterm(2) or mterm(1) or mterm(0);
sig_un <= blk_en and sig_term;
xn(3) <= not(rst_n and x(3));
logterm(0) <= term(0);
logterm(1) <= term(1);
logterm(2) <= term(2);
logterm(3) <= term(3);
logterm(4) <= term(4);
logterm(5) <= term(5);
raw_term(0) <= xn(3) and xn(2) and xn(1) and xn(0);
logterm(6) <= term(6);
logterm(7) <= term(7);
logterm(8) <= term(8);
logterm(9) <= term(9);
logterm(10) <= term(10);
logterm(11) <= term(11);
logterm(12) <= term(12);
logterm(13) <= term(13);
logterm(14) <= term(14);
logterm(15) <= term(15);

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term(0) <= en_term(0) and raw_term(0);

U60 : d_ff
  port map(
    clk => clk0,
    dl => sig_un,
    q => sig_out,
    rst_p => GND
  );

raw_term(1) <= xn(3) and xn(2) and xn(1) and x(0);
term(1) <= en_term(1) and raw_term(1);
raw_term(2) <= xn(3) and xn(2) and x(1) and xn(0);

---- Power , ground assignment ----

GND <= GND_CONSTANT;

end sel_block;

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